GLOBALRAIN

Practices for Secure Software Report

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Document Revision History

Version	Date	Author	Comments
1.0	2/25/26	Hunter Miller	

Client

Instructions

Submit this completed practices for secure software report. Replace the bracketed text with the relevant information. You must document your process for writing secure communications and refactoring code that complies with software security testing protocols.

- Respond to the steps outlined below and include your findings.
- Respond using your own words. You may also choose to include images or supporting materials. If you include them, make certain to insert them in all the relevant locations in the document.
- Refer to the Project Two Guidelines and Rubric for more detailed instructions about each section of the template.

Developer
Hunter Miller

1. Algorithm Cipher

Artemis Financials' goal is to provide financial plans for customers internationally. As a developer, I would recommend using Advanced Encryption Standard (AES-256). AES-256 is a symmetric block cipher that uses a 256-bit key to both encrypt and decrypt data. AES-256 creates 2^{256} possible key combinations, making brute force attacks considered impossible. This level of security ensures that data remains confidential when transmitted locally or internationally.

Discuss the hash functions and bit levels of the cipher – While AES-256 ensures data confidentiality, hash functions such as SHA-256 are used to ensure data integrity. SHA-256 creates a fixed 256-bit value for any input. Hash functions are one-way operations in the sense that data cannot be reconstructed from the hash. SHA-256 is widely used in security applications because it is extremely unlikely for two different inputs to produce the same hash.

2. Certificate Generation
Insert a screenshot below of the CER file.

```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Hunter>"C:\Program Files\Java\jdk-25.0.2\bin\keytool" -printcert -file server.cer
Owner: CN=Hunter Miller, OU=SNHU, O=SNHU, L=Zanesville, ST=OH, C=US
Issuer: CN=Hunter Miller, OU=SNHU, O=SNHU, L=Zanesville, ST=OH, C=US
Serial number: 81935e20379bffde
Valid from: Sat Feb 14 19:27:42 EST 2026 until: Tue Feb 09 19:27:42 EST 2027

Certificate fingerprints:
         SHA1: 58:A3:AB:9D:27:52:E9:3F:BC:F9:6B:7D:61:FA:FE:1D:36:38:53:40
Certificate fingerprints:
Certificate fingerprints:
Certificate fingerprints:
Certificate fingerprints:
         SHA1: 58:A3:AB:9D:27:52:E9:3F:BC:F9:6B:7D:61:FA:FE:1D:36:38:53:40
         SHA256: 3D:53:0B:2F:AF:51:98:3F:DB:18:17:61:97:4D:54:7F:DB:B1:BD:43:86:15:A5:0A:61:E4:4B:07:50:CA:B1:32
Signature algorithm name: SHA384withRSA
Subject Public Key Algorithm: 2048-bit RSA key
Version: 3

Extensions:

#1: ObjectId: 2.5.29.14 Criticality=false
SubjectKeyIdentifier [
KeyIdentifier [
0000: 1B 9F FB 5B 22 A7 9A C4   AB BF 74 31 E8 27 71 82  ...["......t1.'q.
0010: 16 81 02 53                                        ...S
]
]



C:\Users\Hunter>
```

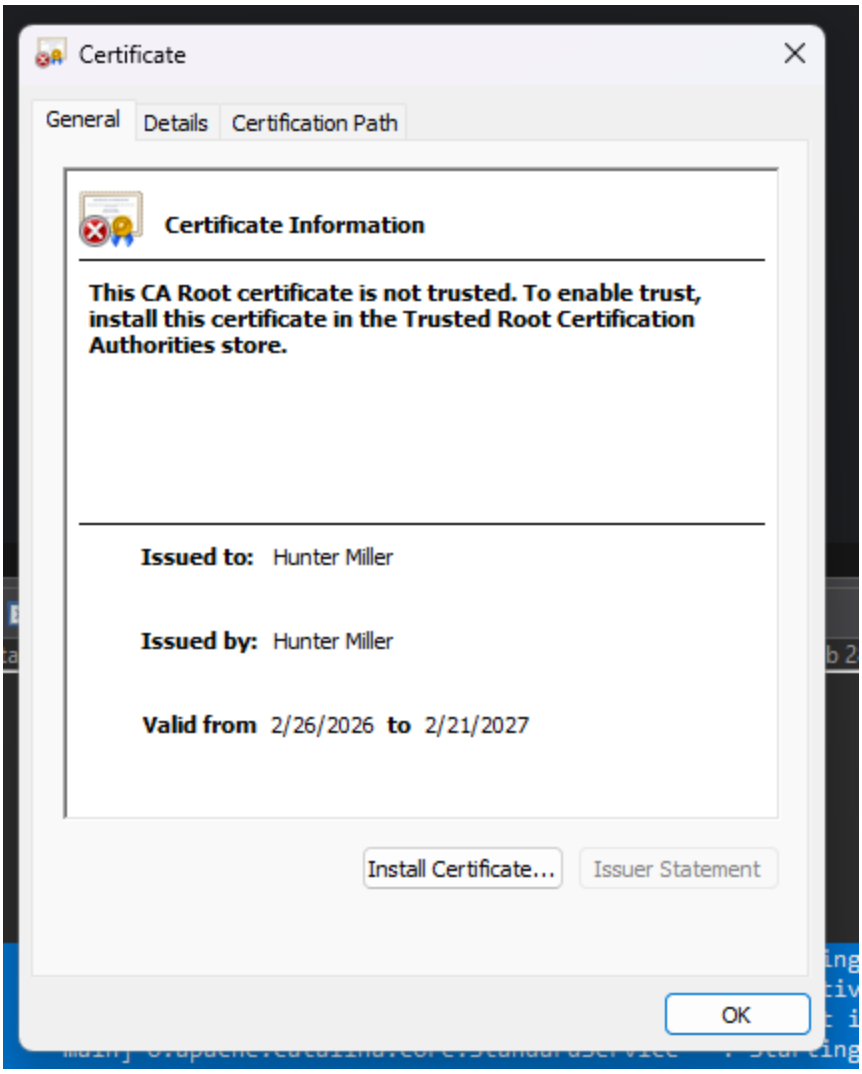
3. Deploy Cipher

Insert a screenshot below of the checksum verification.

SHA-256 Checksum Demo Name: Hunter Miller Unique String: CS305_Project2 Generated Checksum: 318b2c5fff4b3ea4c7afa72a028e5f3a46ab698b6977ca553789d900cfae6d27 Verification Result: true

4. Secure Communications

Insert a screenshot below of the web browser that shows a secure webpage.

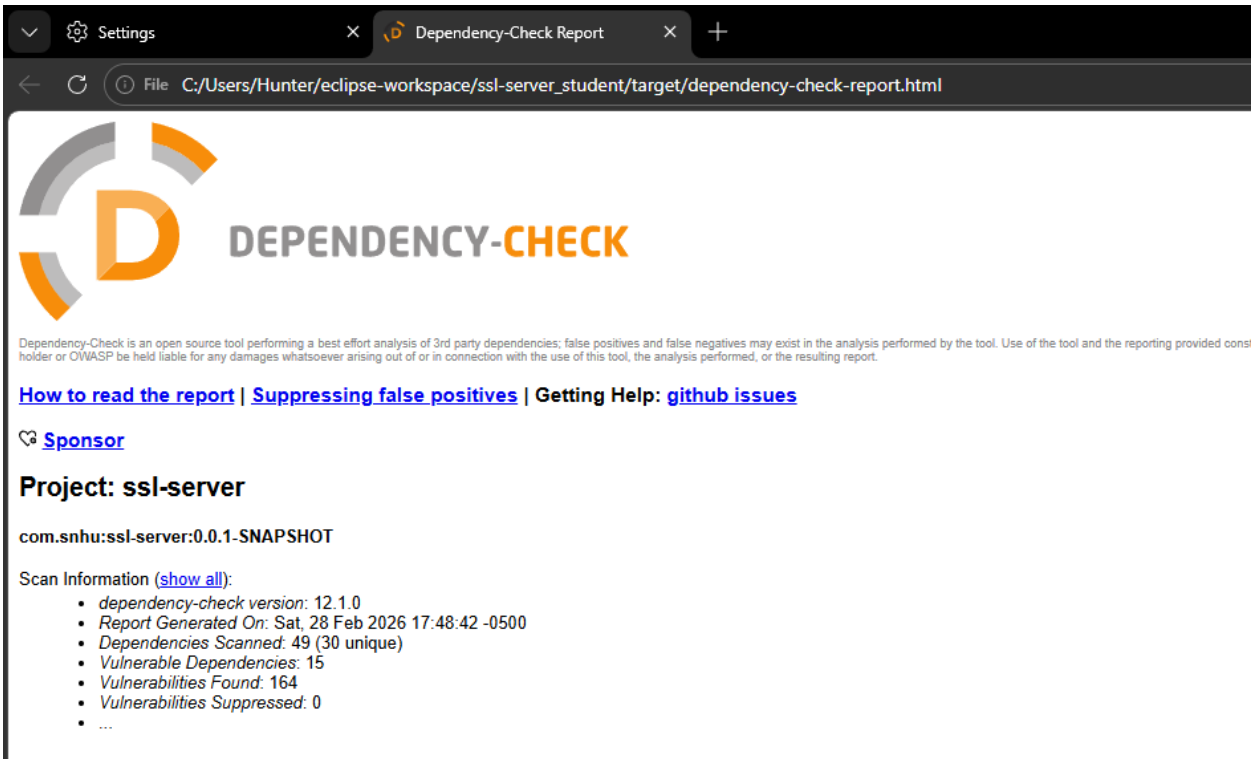


Installed Certificate, the site still says Not Secure.

SHA-256 Checksum Demo Name: Hunter Miller Unique String: CS305_Project2 Generated Checksum: 318b2c5fff4b3ea4c7afa72a028e5f3a46ab698b6977ca553789d900cfae6d27 Verification Result: true

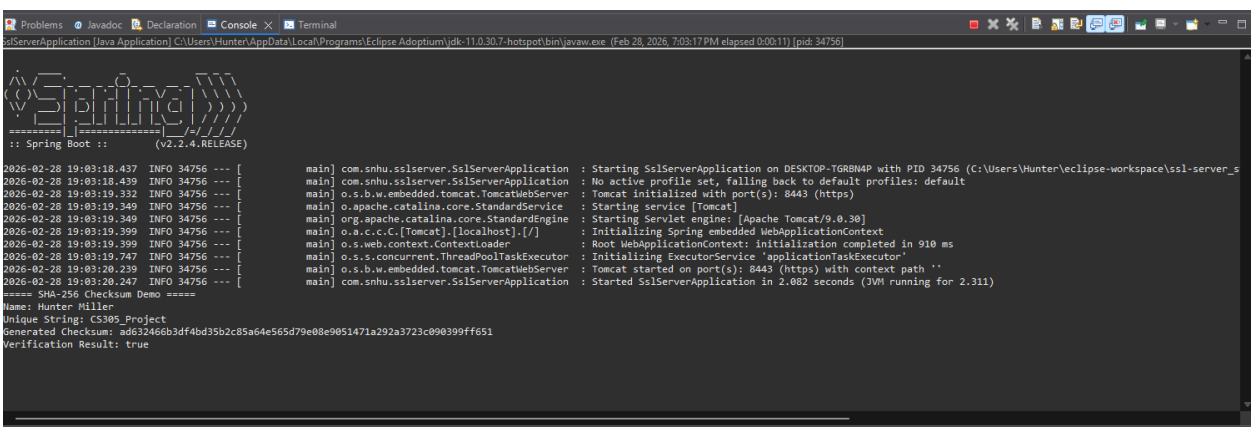
5. Secondary Testing

Insert screenshots below of the refactored code executed without errors and the dependency-check report.



6. Functional Testing

Insert a screenshot below of the refactored code executed without errors.



7. Summary

The refactored SSL application was refactored and executed successfully. The SHA-256 checksum was isolated in the ChecksumVerification. The ChecksumController class handles exceptions to avoid exposing sensitive data. Overall, input data is processed securely and the checksum verification ensures data integrity.

8. Industry Standard Best Practices

By isolating the SHA-256 checksum operations, the application maintains security while reducing the risk of potential vulnerabilities. Input data is processed, and exceptions are handled to prevent data from being exposed. These practices strengthen the overall security and reliability of the application.